

Sihan Chen

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EDUCATION

University of Toronto | CGPA: 3.9/4.0

Sept. 2022 – Present

Bachelor of Applied Science (BASc) in Computer Engineering (AI & Robotics Minor) + PEY Co-Op

Relevant Coursework: Data Structures & Algorithms, Operating System, Introduction to Databases, Linear Algebra

SKILLS AND RELEVANT KNOWLEDGE BACKGROUND

Programming & Frameworks: C/C++, Python, Ansible, JavaScript, React, PostgreSQL

Tools & Technologies: AWS Services, Docker, Kubernetes, Git, Linux (Shell Scripting), Django

EXPERIENCE

Cluster Deployment Software Engineer Intern | Cerebras | Toronto, ON

May 2025 – Present

- Developed and expanded Python-based automation framework supporting multi-vendor server deployment including OS installation, firmware updates, and networking adaptation, reducing manual provisioning effort from days to hours.
- Built Python backend dashboard providing real-time network path validation and automated troubleshooting recommendations, eliminating manual console access for datacenter operators.
- Led two 5MW datacenter deployments and multiple customized cluster labs, fostering cross-functional collaboration with networking and operations teams to drive rapid infrastructure provisioning and accelerate revenue realization.
- Optimized server provisioning workflows using Python and Ansible playbooks through 60+ reliability and process improvement contributions, reducing deployment time by 20%.
- Identified and documented 50+ deployment failure scenarios (including Kubernetes-related issues) with automated recovery solutions, enabling future deployments to self-recover without manual intervention.

Research Intern with Prof. Jonathan Rose | University of Toronto | Toronto, ON

May 2024 – Aug 2024

- Contributed and maintained a full-stack GPT-4-powered chatbot with a React survey frontend and a **Python (Sanic)** backend, using **API Gateway** for handling requests and **ECS** for backend deployment.
- Optimized backend workflows, reducing verbosity in chatbot responses through advanced prompt structuring techniques, achieving a **30% reduction in operational costs** while preserving response accuracy and relevance.
- Co-developed a **modularized, scalable** data generation pipeline in **Python** capable of creating **40000** sentences from any domain using **chain-of-thought** reasoning and iterative model refinement to enhance generation quality.

PROJECTS

Sole Developer | Personal Website

Jul. 2024 – Present

- Developed and deployed a **React**-based blog website (www.sihan.ca) to showcase technical experiences, leveraging AWS service including **Amplify**, **CodePipeline**, and **CodeDeploy** for automated build, test and deployment (**CI/CD**).
- Configured a **Docker** environment to deploy a Content Management System (**CMS**) using **Strapi** and a **Python** backend to organize and manage page data dynamically on an **EC2** instance.
- Integrated **OpenAI APIs** for customized experiences and implementing a **database** to store user inputs and blog posts.

Leader | Software Design and Communication (ECE297) – Day Travel Map

Jan. 2024 – Apr. 2024

- Designed a user-friendly and responsive city mapping GIS for commuters using **C++ STL** and implemented various pathfinding algorithms including **Dijkstra** and **A*** to improve pathfinding efficiency by **50%**.
- Draw components of cities using the **EZGL** library, a wrapper around **GTK**, based on data from OpenStreetMap.

Leader | Fundamentals of Deep Learning (APS360) Course Project

May. 2023 – Aug. 2023

- Led the development of a **Convolutional Neural Network (CNN)** to classify images of 5 car brands from a dataset of augmented surveillance-natured car images, achieving an **82% test accuracy**.
- With a limited dataset, leveraged **PyTorch** for **image augmentation** techniques such as Color Jitter, cropping, resizing, and Gaussian noise to improve the model's performance.